数学I 1学期 復習プリント ① 数と式（展開・因数分解）

3年　組　番　氏名 _____

1　次の式を展開せよ。

(1) $(x+3)(x-7)$

(2) $(2x-5)^2$

(3) $(3x+2y)(3x-2y)$

(4) $(2x+3)(3x-4)$

(5) $(x+2y)(x-5y)$

(6) $(x+y-2)^2$

(7) $(x+1)(x-1)(x^2+1)$

(8) $(x^2-2x+3)(x^2-2x-1)$

2　次の式を因数分解せよ。

(1) $x^2-5x-24$

(2) $3x^2+7x+2$

(3) $4x^2-25y^2$

(4) $3ax^2-12a$

(5) $2x^2+5xy+3y^2$

(6) x^2-y^2+4y-4

(7) x^4-13x^2+36

(8) $ax+by-ay-bx$

3　次の計算をせよ。分母に根号があるときは有理化せよ。

(1) $\sqrt{48}-\sqrt{27}+\sqrt{12}$

(2) $(\sqrt{5}+\sqrt{2})(\sqrt{5}-\sqrt{2})$

(3) $(2\sqrt{3}-\sqrt{2})^2$

(4) $\dfrac{6}{3-\sqrt{3}}$

(5) $\dfrac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}$

(6) $\sqrt{7-4\sqrt{3}}$

4　$x=2+\sqrt{3}$、$y=2-\sqrt{3}$ のとき、次の値を求めよ。

(1) $x+y$、xy

(2) x^2+y^2

(3) x^3+y^3

(4) $\dfrac{y}{x}+\dfrac{x}{y}$

5　次の不等式を解け。

(1) $5x - 3 < 2x + 9$

(2) $\dfrac{x-2}{3} - \dfrac{2x+1}{4} \geqq 1$

(3) $\begin{cases} 3x - 1 \leqq x + 5 \\ 2x + 3 > x - 1 \end{cases}$

(4) $0.3x - 0.5 < 0.1x + 0.7$

(5) $-1 < 2x - 5 \leqq 3$

(6) $2(x - 3) > 5x + 3$

(7) $\dfrac{x+1}{2} - \dfrac{2x-1}{3} < 1$

(8) $\begin{cases} 2x + 1 < x + 4 \\ 5x - 3 \geqq x + 1 \end{cases}$

(9) $0.5x - 1.2 \geqq 0.2x + 0.3$

(10) $-3 \leqq 4 - x < 6$

6　次の2次関数を $y = a(x-p)^2 + q$ の形に変形し、頂点の座標と軸を答えよ。

(1) $y = x^2 - 6x + 5$

(2) $y = x^2 + 3x + 1$

(3) $y = 2x^2 - 8x + 3$

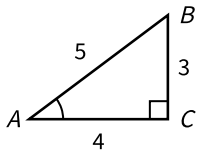
(4) $y = -3x^2 + 6x + 1$

(5) $y = \frac{1}{2}x^2 + 2x - 1$

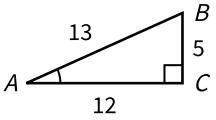
(6) $y = -x^2 + 5x$

7　次の直角三角形について、$\sin A$、$\cos A$、$\tan A$ の値を求めよ。

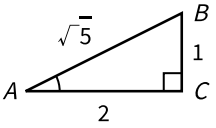
(1)



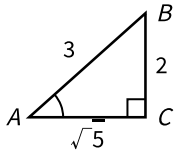
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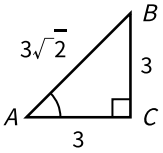
(3)



(4)



(5)



(6)

